Physical Geography: Tropical Monsoons & Atmospheric Circulation

1. Differential Thermal Heating of Landmass and Ocean

Monsoons represent large-scale seasonal reversals in atmospheric surface winds caused by the differential specific heat capacity between the Asian landmass and the Indian Ocean.

2. The Intertropical Convergence Zone (ITCZ) & Coriolis Deflection

During northern hemisphere summer, intense heating of the Tibetan Plateau generates a profound low-pressure thermal trough, drawing moist maritime air northwards across the equator, deflected rightwards by the Coriolis effect.